

The exact nine-word boundary—and what fails outside it

A uniform one-step square deletion exists iff $H_{4K(u)} = \{1\}$ iff $u \mid 36$.

All and only the automatic words

$u = 1$ $K(u) = 1, m = 4$ $H_m = \{1\}$	$u = 2$ $K(u) = 2, m = 8$ $H_m = \{1\}$	$u = 3$ $K(u) = 3, m = 12$ $H_m = \{1\}$
$u = 4$ $K(u) = 2, m = 8$ $H_m = \{1\}$	$u = 6$ $K(u) = 6, m = 24$ $H_m = \{1\}$	$u = 9$ $K(u) = 3, m = 12$ $H_m = \{1\}$
$u = 12$ $K(u) = 6, m = 24$ $H_m = \{1\}$	$u = 18$ $K(u) = 6, m = 24$ $H_m = \{1\}$	$u = 36$ $K(u) = 6, m = 24$ $H_m = \{1\}$

For every proper source at these words, $q_0 = K(d)$ gives $k = q_0^2$ and $R' = R/q_0^2$.

The complete square family has $\tau(t/q_0)$ distinct targets; no source-existence claim is added.

Outside boundary: the exact $u = 8$ obstruction

$$\begin{aligned} m &= 16, \quad H_{16} = \{1, 9\}, \quad b = 5 \\ \delta &= 71, t = 13, R = \delta t^2 = 11999 \\ p &= 48116881, a = 12032220, d = t = 13 \end{aligned}$$

Every possible same-word target residual fails:

$$\begin{aligned} R' &= \delta = 71 \\ p + R' &\equiv 8 \equiv 0 \pmod{16} \end{aligned}$$

$$\begin{aligned} R' &= \delta t = 923 \\ p + R' &\equiv 12 \equiv 0 \pmod{16} \end{aligned}$$

$$\begin{aligned} &\text{middle complement } u^* = a^2/u \\ 4K(u^*) &= 24064440 > R; \text{ its exact denominator is still } 13 \end{aligned}$$

This blocks the residual-divisor method, not Erdős-Straus.

A separate endpoint state (13122786, 631427532350466, 144350643) has reciprocal sum $4/48116881$.